

WORKSHOP BENCH PROJECT SUITE

THE BENCH — 35 SMALL WEB TOOLS

Complete Project Overview & Comprehensive Implementation Blueprint (35 Applications)

PROJECT SUMMARY & STATS

- Total Tools: 35 Fully Implemented Web Apps
- Infrastructure: 0 Servers / 0 Build Steps / 100% Client-Side LocalStorage
- Stack: HTML5, CSS3, ES6+ JS
- Attribution: Built by Vishu — 'A bench of 35 small tools, made to be used, not just looked at.'

1. What This Is & Design Philosophy

"The Bench" (project folder: 20-web-projects) is a static, no-build, no-backend collection of 35 self-contained web applications hung on a single hub page styled like a workshop pegboard. Every tool is built with plain HTML, CSS, and JavaScript without frameworks or bundlers. Data persistence relies entirely on the browser's localStorage under namespaced keys.

Core Engineering & Design Principles

- **Zero Infrastructure:** No node server, no build step, and no package manager required. Opening index.html directly in a browser unlocks 100% functionality.
- **One Unified Visual Shell:** Every tool page shares the same header navigation (<- The Bench), consistent typography, button chips, glass-panel styling, and dark/light theme toggles.
- **Real Logic over Stubs:** Every app implements complete domain logic (e.g., scientific calculator mode, state-wise CGST/SGST/IGST tax splits, and Breadth-First Search dungeon solvability verification).
- **Namespaced Local Storage:** Data persistence (saved tasks, high scores, color palettes, invoice counts) utilizes Bench.Storage to prevent key collisions across tools.
- **Graceful Network Degradation:** All external API calls wrap in try/catch blocks with toast fallback alerts to guarantee offline stability.

2. Project Structure & Code Architecture

WORKSPACE FILE ARCHITECTURE

```
20-web-projects/
├── index.html          <- Hub Dashboard ("The Bench")
├── assets/
│   ├── css/
│   │   ├── style.css   <- Core Design System & Tokens (Colors, Type, Themes)
│   │   ├── loader.css  <- CSS Loading Animations & Skeleton Shimmers
│   │   └── parallax.css <- Parallax Hero Background Scroller
│   └── js/
│       ├── storage.js  <- Bench.Storage, Bench.toast, Bench.debounce
│       └── main.js      <- Catalog Data, Search, Filter & GSAP Animations
└── projects/
    ├── 01-todo-list/ ... 35-expense-invoice-tracker/
    └── [Each project contains exactly: index.html, script.js, style.css]
```

3. The Complete Catalog of 35 Web Tools

#	Tool Title	Category	Key Features & Highlights
01	Task Board	Productivity	Categorized tasks, priority tags, drag-to-reorder, saved filters
02	Calculator	Utility	Standard and scientific modes with a running history drawer
03	Snake	Game	Canvas arcade snake with skins, sound toggle, local high-score table
04	Sign-Up Flow	Form	Multi-step registration with live validation and password-strength meter
05	Hero Slider	Media	Touch-swipe slider with autoplay, captions, thumbnail rail
06	BMI Gauge	Utility	Interactive sliders feeding a live gauge and saved history log

07	Password Smith	Utility	Entropy-rated password generator with a small saved vault
08	Palette Forge	Utility	Lock colors, convert HEX/RGB/HSL, export a saved palette
09	Counter Bank	Productivity	Multiple named counters with step size, targets, and history
10	QR Bench	Utility	Custom QR codes with color/size control and a scan history log
11	Weather Bench	Data	City search with a 5-day outlook and saved favorite locations
12	Ledger	Finance	Income vs. expense bars, category filters, monthly summaries
13	Quiz Master	Game	Timed trivia with card-flip reveals and a top-scores board
14	Focus Timer	Productivity	Configurable work/break cycles with streak tracking
15	Memory Flip	Game	3D flipping match game across three grid sizes with a move counter
16	Markdown Bench	Productivity	Live-preview note editor with tags, word count, and export
17	Typing Trial	Game	WPM and accuracy trainer with a live results graph
18	Recipe Box	Data	Ingredient search, cook timers, bookmarked favorites
19	Unit Bench	Utility	Currency, length, weight, temperature, and speed conversions
20	Sound Deck	Media	Playlist player with an animated visualizer and equalizer
21	Car Racing	Game	Lane-dodging racer that speeds up the longer

			you survive
22	Treasure Hunter	Game	Procedural dungeon crawl — BFS verified solvable layout
23	Space Defender	Game	Wave-based invader shooter with escalating enemy fire
24	Zombie Arena	Game	Twin-stick survival against endless waves of the horde
25	Tower Defense	Game	Place towers, hold the path, survive escalating waves
26	Gradient Generator	Creative	Color-theory palettes turned into ready CSS gradients
27	Logo Maker	Creative	Emblem, name, and layout combined into an exportable logo
28	Poster Designer	Creative	Drag-to-place headline and subtitle over custom background
29	Pixel Art Studio	Creative	Grid-based pixel painter with fill tool and PNG export
30	Social Creator	Creative	Square, story, or landscape posts ready to export
31	Receipt Maker	Business	Itemized receipts with tax, ready to print or save as PDF
32	Invoice Generator	Business	Client invoices with line items, tax, and due dates
33	GST Billing	Business	Indian GST invoices with automatic CGST/SGST/IGST split
34	Quotation Maker	Business	Client estimates with validity dates and clear terms
35	Expense Tracker	Business	Invoiced, collected, pending, and expenses in one dashboard

4. Full Implementation Blueprint: All 35 Core Projects

Below is the complete implementation spec for all 35 web applications generated in the project suite:

- **01. Task Board:** Task management with drag & drop reordering, priority badges (High, Medium, Low), search/category filters, and state saved in Bench.Storage.
- **02. Calculator:** Dual-mode calculator (Standard & Scientific), trigonometric & logarithmic functions, slide-out calculation history drawer, keyboard shortcuts.
- **03. Snake Game:** Classic HTML5 Canvas arcade snake with smooth grid movement, retro skins, audio oscillator effects, local high-score table.
- **04. Registration Form:** Interactive multi-step sign-up flow with real-time field validation, password entropy strength meter, live preview modal, auto-draft saving.
- **05. Image Slider:** Parallax hero image carousel with GSAP split-text captions, touch/swipe navigation, thumbnail rail, pause-on-hover autoplay.
- **06. BMI Calculator:** Interactive weight/height visual sliders feeding a real-time BMI gauge, health recommendation categories, historical progress log.
- **07. Password Generator:** Customizable security criteria (Length, Uppercase, Numbers, Symbols), entropy strength indicator, click-to-copy, local password vault.
- **08. Color Generator:** Palette & gradient generator, color locking, HEX/RGB/HSL conversion, palette export, saved favorites in LocalStorage.
- **09. Counter App:** Multi-counter dashboard, customizable step increments, max/min target limits, count history log, persistent counter values.
- **10. QR Code Generator:** Dynamic QR generator, background/foreground color pickers, size controls, PNG/SVG download support, scan history log.
- **11. Weather App:** Weather dashboard with city search, animated weather status cards, 5-day forecast, favorite cities saved in LocalStorage.
- **12. Expense Tracker:** Personal financial ledger with income vs. expense visual distribution bars, category filtering, monthly expense summaries.
- **13. Quiz Master:** Timed trivia challenge across multiple categories, GSAP 3D card-flip answer reveals, live countdown timer, top score leaderboards.
- **14. Pomodoro Focus Timer:** Customizable work/break timer cycles, ambient focus audio generators, session streak counter, task linking.
- **15. Memory Matching Game:** 3D card flipping memory game with 3 grid sizes (4x4, 6x6, 8x8), move counter, timer, local best-time leaderboards.
- **16. Markdown Notes App:** Rich Markdown editor with live split-screen preview, tag organization, search filter, text export, auto-save persistence.
- **17. Typing Speed Test:** Real-time WPM & accuracy measurement, visual text highlighting, mistake tracking, interactive WPM performance graph chart.

- **18. Recipe Finder:** Culinary recipe box with ingredient search, integrated cooking step timers, calorie info cards, favorite bookmarking.
- **19. Unit Converter:** Multi-category converter (Currency, Length, Weight, Temp, Speed) with instant bi-directional calculation and conversion history.
- **20. Music Player UI:** Custom audio player featuring GSAP animated sound visualizer bars, playlist controls, seek bar, equalizer presets.
- **21. Car Racing Game:** Lane-dodging arcade racer with dynamic speed escalation and obstacle collision detection.
- **22. Treasure Hunter Adventure:** Procedural dungeon crawl — BFS algorithm verifies map solvability before rendering.
- **23. Space Defender:** Wave-based arcade space shooter with enemy projectiles, particle explosion FX, score tracking.
- **24. Zombie Survival Arena:** Twin-stick survival shooter game against endless zombie waves, ammo pickups, high-score table.
- **25. Tower Defense:** Tactical tower placement defense game with enemy pathfinding, upgrade trees, escalating wave system.
- **26. AI Gradient Generator:** Deterministic HSL color-theory math engine generating CSS linear/radial gradients with copyable CSS code.
- **27. Logo Maker:** Canvas emblem designer with icon selection, custom typography, shadow effects, and PNG/SVG export.
- **28. Poster Designer:** Interactive headline/subtitle poster layout editor with image background upload and drag-and-drop text positioning.
- **29. Pixel Art Studio:** Grid canvas pixel painting tool with color palette picker, bucket fill tool, eraser, and PNG export.
- **30. Social Media Post Creator:** Multi-format canvas maker for Instagram Square (1:1), Story (9:16), or Landscape (16:9) post templates.
- **31. Receipt Generator:** Itemized transaction receipt builder with sales tax split, subtotal calculations, and print-to-PDF layout.
- **32. Invoice Generator:** Client invoice generator with line items, tax rate, payment terms, and native browser PDF print styling.
- **33. GST Billing System:** Indian GST invoice software with auto intra-state (CGST + SGST) vs inter-state (IGST) tax calculation splits.
- **34. Quotation & Estimate Maker:** Client proposal and estimate creator with validity date tracking, discount calculations, and terms text.
- **35. Expense & Invoice Tracker:** Financial dashboard displaying total invoiced, collected revenues, pending receivables, and expense breakdowns.